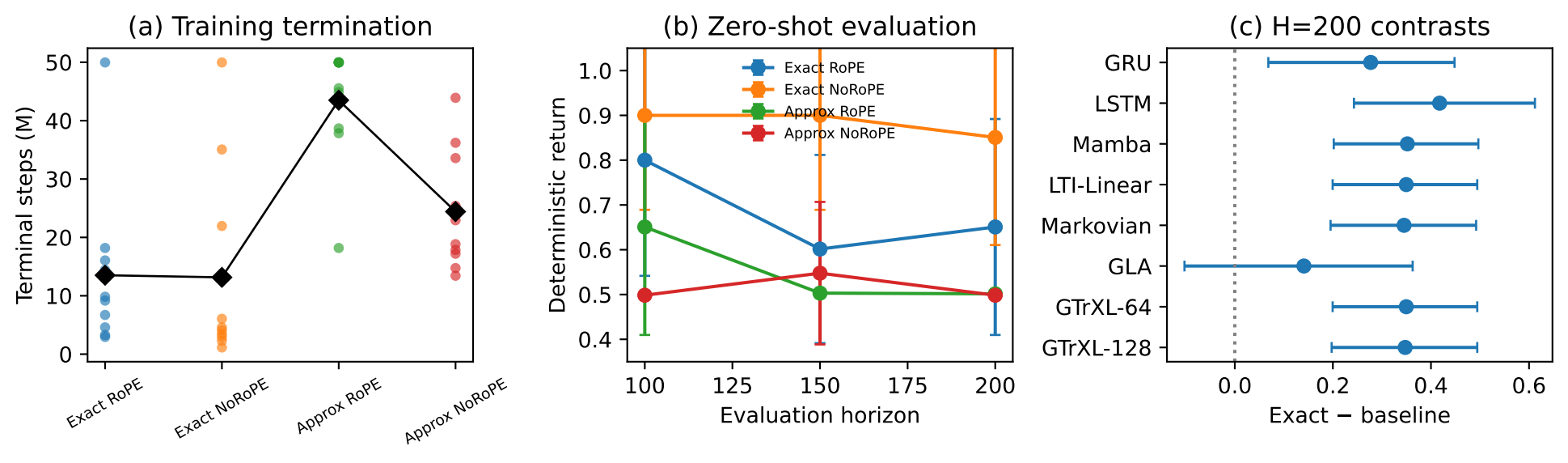
(a) Training termination
Terminal steps (M)
0
10
20
30
40
50
Exact RoPE
Exact NoRoPE
Approx RoPE
Approx NoRoPE
(b) Zero-shot evaluation
Deterministic return
0.4
0.5
0.6
0.7
0.8
0.9
1.0
100
125
150
175
200
Evaluation horizon
Exact RoPE
Exact NoRoPE
Approx RoPE
Approx NoRoPE
(c) H=200 contrasts
GRU
LSTM
Mamba
LTI-Linear
Markovian
GLA
GTrXL-64
GTrXL-128
0.0
0.2
0.4
0.6
Exact − baseline